A Hyperparameters

Table A1 lists the tested values. Bold marks the paper's default.

Table A1: Hyperparameters and values tested in the sensitivity sweeps. Bolded values are defaults.

Stage	Symbol	Meaning	Tested values
Sampling	$p_\star$	selectivity threshold	0.01, 0.02, **0.05**, 0.1, 0.2
	b_0	batch size	8, 16, 32, 64, 128
	μ	cluster penalty	0.5, 1, **1.5**, 2.5, 4
	$I_{\max}$	maximum cluster count	1, 2, 4, **6**, 8
	$\varepsilon_{\text{explore}}$	exploration share	0, 0.1, **0.25**, 0.5
	f_{pilot}	pilot share	0.05, 0.1, 0.2, **0.3**
Graph refinement	k	neighbors per row in the k-NN graph	5, 10, 30, 50
	β	weight of bias diffusion	0, 0.2, 0.4, 0.6, **0.8**, 1
	γ	weight of homogeneity propagation	0, 0.2, 0.4, 0.6, **0.8**, 1
	θ_{ref}	threshold for refined labels	0.3, 0.4, **0.5**, 0.6, 0.7
Calibration	κ_L	share of sampling budget for proxy-training labels	**0.6**, 0.7, 0.8, 0.9
	R_{fold}	upper bound on cross-fitting folds	2, 3, **5**, 10

B Sensitivity Studies

Setup. Figure A1 reports geometric-mean F_1 and latency over subsets of the 14 queries in Table 2 of the main paper.

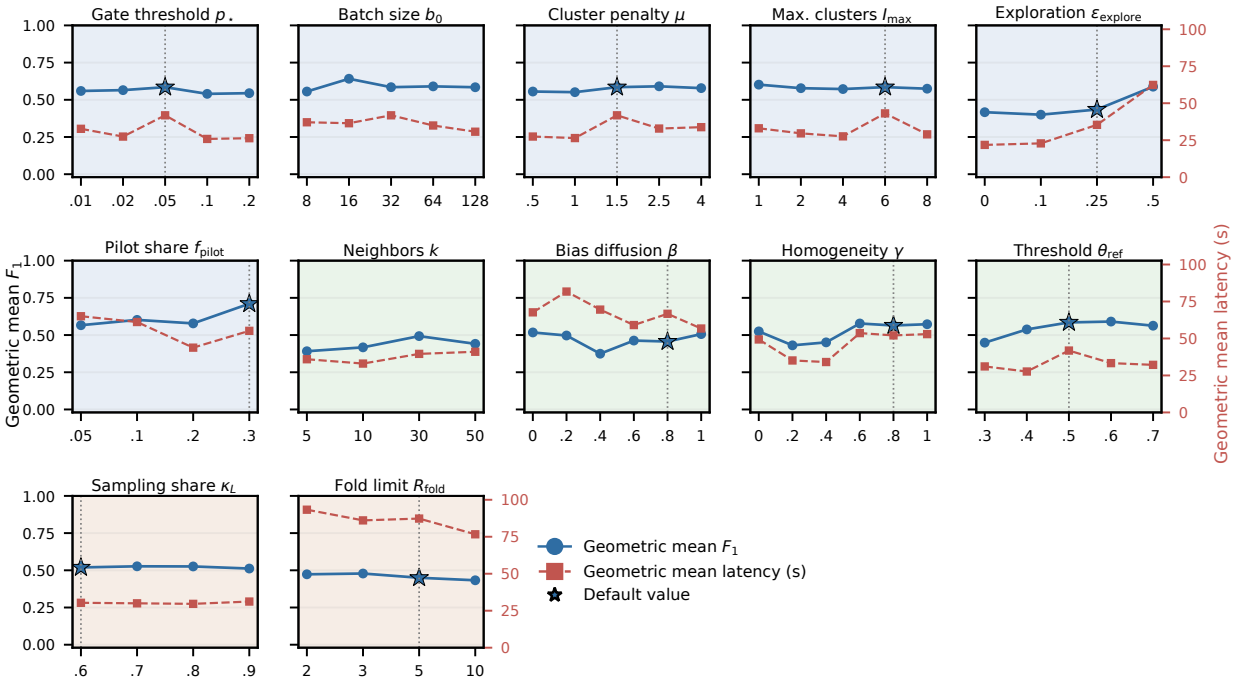


Figure A1: Geometric-mean F_1 (solid) and latency (dashed). Stars mark the default values.

Sampling. Varying $p_\star$, μ, or $I_{\max}$ changes F_1 by less than 0.05 within each sweep. A batch size of 16 reaches 0.641, compared with 0.585 at 32. A pilot share of 0.3 gives the highest F_1 in its sweep, at 0.711.

Graph refinement. Among the tested neighbor counts, $k = 30$ gives the highest F_1. Increasing θ_{ref} from 0.3 to 0.6 raises F_1 from 0.449 to 0.591. The results vary non-monotonically with β and γ.

Calibration. Varying κ_L changes F_1 by less than 0.02. Increasing the fold limit from 3 to 10 reduces F_1 from 0.478 to 0.433.